PREDICTING E-COMMERCE PURCHASE LIKELIHOOD

Business Recommendations Report

Derived from: purchase_prediction_analysis.ipynb

Dataset: e_commerce_customer_data.csv (5,000 customers)

Selected Model: Optimized Logistic Regression

Executive Summary

This report translates the findings of a machine-learning analysis of 5,000 e-commerce customer sessions into concrete, actionable business recommendations. The underlying project built and compared three classification models — Logistic Regression, Decision Tree, and Random Forest — to predict whether a website visitor will complete a purchase, using behavioral and demographic signals such as pages viewed, time on site, cart activity, prior purchase history, discount usage, device type, and traffic source.

After baseline training and hyperparameter optimization via GridSearchCV (5-fold cross-validation, optimized for F1-score), the Logistic Regression model was selected as the production model, delivering 84.5% accuracy, 85.4% precision, 83.2% recall, an F1-score of 0.843, and a ROC-AUC of 0.922 on a held-out test set of 1,000 customers. This level of performance is strong enough to support real-time scoring of customer purchase intent and to power targeted marketing, personalization, and retention initiatives.

The analysis surfaced a clear, actionable hierarchy of purchase drivers. A customer's prior purchase history and current cart activity together account for the majority of the model's predictive power, followed by session engagement (time on site, pages viewed) and discount usage. Demographic factors such as age and gender, along with acquisition channel, contribute only marginally. This has direct implications for where the business should invest its marketing, personalization, and retention budget.

The recommendations that follow are organized by business function — Marketing & Acquisition, On-Site Personalization & UX, Loyalty & Retention, Pricing & Promotions, and Data & Analytics Operations — so that each team can act on the findings that are most relevant to them. A phased implementation roadmap and a summary of model limitations close out the report.

1. Business Context and Objectives

E-commerce businesses operate in an environment where the overwhelming majority of site visits do not convert into purchases. Marketing spend, personalization budgets, and customer-service attention are finite resources, and the ability to distinguish a visitor who is likely to buy from one who is not is directly tied to revenue efficiency. The purpose of this project was to build a data-driven system that scores purchase likelihood for individual customer sessions, enabling the business to act differently depending on that score — for example, by serving a discount to a hesitant high-intent shopper, or by suppressing costly retargeting spend on a visitor unlikely to convert regardless of incentive.

The specific business objectives behind this analysis were to:

- Quantify which customer behaviors and attributes are most predictive of purchase completion.
- Build a classification model accurate enough to be trusted for operational decisions such as discount targeting and retargeting suppression.
- Segment customers into Low, Medium, and High purchase-likelihood tiers to support tiered marketing and service treatment.
- Provide a foundation (a saved, reusable model artifact) that can be integrated into live systems such as the website, CRM, or ad platform.

The resulting model and the customer-level purchase-probability scores it generates (available in the accompanying scored dataset) are intended to be consumed by Marketing, Growth, Product, and CRM teams as an operational input rather than a one-off research exercise.

2. Data Overview

The analysis used a dataset of 5,000 unique customers, each described by 12 attributes capturing demographics, session behavior, and purchase outcome. The data was clean at the outset — there were no missing values and no duplicate records — and the target variable (Purchase) was perfectly balanced, with exactly 2,500 customers who purchased and 2,500 who did not. This balance is a favorable characteristic for classification modeling, as it removes the class-imbalance problems that often complicate purchase-prediction tasks in practice and means accuracy is a meaningful metric rather than a misleading one.

2.1 Fields Used

CustomerID	Unique customer identifier (excluded from modeling features)
Age	Customer age, 18–59 years
Gender	Male / Female
DeviceType	Mobile, Desktop, or Tablet
TrafficSource	Email, Social Media, Search Engine, or Direct
PagesViewed	Number of pages viewed in the session (1–14)
TimeOnSite	Minutes spent on site (1–9)
CartItems	Number of items added to cart (0–5)
PreviousPurchases	Count of prior completed purchases (0–9)
AverageOrderValue	Historical average order value in currency units (20–299)
DiscountUsed	Whether a discount code was applied (0/1)
Purchase	Target variable: did the customer purchase (1) or not (0)

Before modeling, numeric fields were imputed (mean) and standardized, and categorical fields were imputed (most frequent) and one-hot encoded, using a scikit-learn ColumnTransformer pipeline. This ensures the same preprocessing is applied consistently whether the model is used in training, testing, or live production scoring.

3. Methodology Summary

Three candidate algorithms were trained on an 80/20 stratified train-test split (4,000 training / 1,000 test customers, random_state=42), each wrapped in an identical preprocessing pipeline so that model performance differences reflect the algorithm, not data handling inconsistencies:

1. Logistic Regression — a linear model prized for interpretability and stability, and a strong baseline for binary outcomes like purchase/no-purchase.
2. Decision Tree — a single tree-based model capable of capturing non-linear interactions, but prone to overfitting without constraints.
3. Random Forest — an ensemble of many decision trees, generally more robust than a single tree and useful here for its interpretable feature-importance output.

All three models were first evaluated at default ("baseline") settings, then each was tuned using GridSearchCV with 5-fold cross-validation, optimizing for F1-score rather than raw accuracy — a deliberate choice that balances precision and recall rather than letting the model favor whichever class is easier to predict. The Logistic Regression search varied regularization strength and solver; the Decision Tree search varied tree depth and split/leaf constraints; the Random Forest search varied the number of trees, depth, and feature sampling strategy.

Based on comparative results (detailed in Section 4), the optimized Logistic Regression was selected as the final production model and persisted to disk for reuse. A supplementary scoring step also converted the model's raw probability output into three business-friendly tiers — Low, Medium, and High purchase likelihood — using probability cutoffs of 0.30 and 0.60, and demonstrated an adjustable decision threshold (0.40) to illustrate how the precision/recall trade-off can be tuned to business needs.

4. Model Performance Summary

The table below summarizes accuracy, precision, recall, F1-score, and ROC-AUC for all three algorithms at both baseline and optimized settings. The selected production model is highlighted.

Model	Status	Optimization Detail	Accuracy	Precision	Recall	F1
Logistic Regression	Baseline	Default sklearn params	0.839	0.848	0.826	0.8
Decision Tree	Baseline	Default sklearn params	0.744	0.761	0.712	0.7
Random Forest	Baseline	Default sklearn params	0.836	0.853	0.812	0.8
Logistic Regression	**Selected / Optimized**	GridSearchCV: C=0.01, penalty=l2, solver=liblinear	**0.845**	**0.854**	**0.832**	**0.8**

The optimized Logistic Regression delivered the best overall balance of accuracy (84.5%), precision (85.4%), and F1-score (0.843) among all six model/status combinations tested, which is why it was chosen as the production model. Random Forest was a close second on baseline accuracy (83.6%) but its optimized configuration (tuned for generalization via limited tree depth) slightly reduced accuracy to 83.4% while keeping performance stable — a reasonable trade against overfitting risk. The Decision Tree consistently underperformed both other approaches, reflecting the tendency of a single tree to either overfit (baseline: 74.4% accuracy) or lose sharpness once regularized (optimized: 78.9% accuracy).

One technical caveat for the data science team: the ROC-AUC values reported in the original analysis for the Decision Tree and Random Forest were computed from the Logistic Regression's predicted probabilities rather than each model's own probability output, so the identical 0.922 ROC-AUC shown for all three models in the raw notebook output should not be read as a true multi-model comparison — only the Logistic Regression's ROC-AUC of 0.922 is confirmed accurate. This does not affect the model selection decision, which was based on accuracy, precision, recall, and F1, but it should be corrected before this figure is quoted externally.

5. Key Business Insights from the Data

Beyond model accuracy, the exploratory analysis and feature-importance results reveal patterns in customer behavior that are directly actionable for the business, independent of which algorithm is used to operationalize them.

5.1 Purchase History Is the Single Strongest Signal

A customer's count of previous purchases was by far the most influential predictor of a new purchase, accounting for roughly 46% of the Random Forest's total decision-making weight. In practice, this means repeat customers are dramatically more predictable — and more likely — to buy again than first-time visitors. The business should treat returning customers as its highest-leverage, lowest-risk segment for upsell and cross-sell campaigns.

5.2 Active Cart Engagement Signals Near-Term Intent

The number of items placed in the cart was the second most important feature (about 24% of importance), and the boxplot analysis in the notebook shows a clear separation: customers who purchase carry visibly more items in their cart than those who do not. This confirms that cart size is a reliable, real-time signal of purchase intent and should be monitored continuously during a live session, not just at checkout.

5.3 Session Engagement (Time and Depth) Matters, but Less Than Commitment Signals

Time on site (about 14% importance) and pages viewed (about 8% importance) both positively correlate with purchase likelihood, but they are secondary to purchase history and cart activity. This suggests that simply keeping a visitor engaged longer is a weaker lever than getting them to actually add items to their cart — engagement should be viewed as a means to drive cart activity, not an end goal in itself.

5.4 Discounts Move the Needle, but Modestly

Discount usage carried roughly 3% importance and the crosstab analysis shows a visible lift in purchase rate among customers who used a discount code. This is a real but comparatively small effect versus behavioral signals, implying that indiscriminate blanket discounting is unlikely to be the most efficient use of promotional budget — discounts appear most effective when targeted at customers who are already showing behavioral signs of intent (high cart activity, high engagement) but have not yet converted.

5.5 Demographics and Acquisition Channel Are Weak Predictors

Age, gender, and traffic source collectively accounted for well under 5% of the model's predictive power. While device type and traffic source showed some visible differences in purchase rate in the exploratory bar charts, these differences were not strong enough to earn meaningful weight in the model. This is an important, if slightly counterintuitive, finding: campaigns segmented purely by demographic or channel are far less efficient than campaigns segmented by behavior (purchase history, cart activity, engagement).

6. Business Recommendations

The following recommendations are grouped by business function so that each stakeholder team can prioritize the actions most relevant to their remit. Recommendations are derived directly from the feature-importance ranking and the model's demonstrated ability to separate likely buyers from unlikely ones.

6.1 Marketing & Customer Acquisition

4. Reallocate acquisition and retargeting budget away from broad demographic or channel-based segmentation and toward behavior-based segmentation, since age, gender, device, and traffic source explain very little of purchase likelihood on their own.
5. Build a "win-back" campaign specifically for lapsed repeat customers (customers with PreviousPurchases > 0 who have not purchased recently), since purchase history is the single strongest

predictor of future purchase — this segment should convert at meaningfully higher rates than cold acquisition traffic.

6. Deprioritize spend on visitors who show low engagement (low TimeOnSite, low PagesViewed) and no cart activity, since the model shows these visitors are unlikely to convert regardless of channel.
7. Track a purchase-probability score per acquisition channel over time to identify whether certain channels (e.g., Search Engine, which showed a small independent feature-importance contribution) are bringing in progressively higher- or lower-intent traffic.

6.2 On-Site Personalization and UX

8. Instrument real-time cart-abandonment triggers: since cart size is the second-strongest predictor of purchase, customers who add items but stall should be flagged for an immediate, low-friction nudge (e.g., a reminder banner, a limited-time incentive, or simplified checkout) before they leave the session.
9. Use the model's purchase-probability score to dynamically adjust the on-site experience — for example, showing trust signals, shipping-cost transparency, or social proof to Medium-likelihood shoppers who may be on the fence, while leaving High-likelihood shoppers' checkout flow undisturbed to avoid introducing friction.
10. Since engagement depth (time on site, pages viewed) matters but is secondary to cart activity, prioritize UX investments that convert browsing into cart additions (e.g., prominent "add to cart" placement, simplified product comparison) over investments purely aimed at increasing dwell time.

6.3 Loyalty, CRM, and Retention

11. Formalize a loyalty or repeat-purchase program: given that prior purchase count is overwhelmingly the top predictor, incremental investment in converting first-time buyers into second-time buyers likely has the highest long-term return of any lever available to the business.
12. Prioritize CRM outreach (email, app notifications, personalized offers) to customers who match the profile of the model's High-likelihood tier but have gone quiet, since they are the most efficient re-engagement targets.
13. Set up an early-warning segment for new customers (PreviousPurchases = 0) with high cart and engagement activity but no completed purchase — this is the group most valuable to convert into a first purchase, since it seeds all future purchase-history value.

6.4 Pricing and Promotions

14. Target discount codes at customers who show high engagement and cart activity but a middling purchase-probability score (the "Medium" tier), rather than distributing discounts broadly, since discount usage's modest overall importance suggests it works best as a targeted nudge rather than a blanket lever.
15. Test whether the discount effect is stronger for new customers than repeat customers (or vice versa); if repeat customers convert reliably without a discount, funds can be redirected to acquisition-stage incentives instead.
16. Avoid using average order value as a primary trigger for discounting — it carried very low feature importance (about 1.5%), indicating it is a weak standalone signal of purchase likelihood.

6.5 Data, Analytics, and Model Operations

17. Deploy the saved Logistic Regression model (log_model.pkl) behind a scoring API or batch job so that Marketing, CRM, and Product teams can consume live purchase-probability scores rather than working from static exports.
18. Correct the ROC-AUC calculation for the Decision Tree and Random Forest models (each should use its own predict_proba output) before those figures are used in any further reporting or model-selection decisions.
19. Revisit the classification threshold periodically: the notebook demonstrates that adjusting the decision cutoff from the default 0.50 to 0.40 changes which customers are flagged as likely purchasers — the optimal threshold should be set based on the relative business cost of a missed sale (false negative) versus a wasted incentive (false positive), not left at the statistical default.
20. Expand the feature set in future iterations to include richer behavioral signals not present in this dataset — such as browsing recency, product category viewed, referral campaign ID, and cart abandonment history — since the current top features (purchase history, cart size) are strong but relatively coarse.
21. Re-run feature importance analysis directly on the selected Logistic Regression model (via its coefficients) in addition to the Random Forest-derived importances used in this analysis, since the two algorithms can weight features differently and the production model's own logic should be the primary reference for business decisions.

7. Risks and Limitations

Several limitations should be kept in mind when operationalizing this model:

- The training data is synthetic or heavily simplified relative to real e-commerce traffic (e.g., TimeOnSite is capped at 9 minutes and PagesViewed at 14), so absolute performance figures may not transfer directly to production traffic with more extreme or noisy behavior patterns; the model should be revalidated against live data before full deployment.
- Class balance in this dataset (50/50 purchase split) is unusually favorable and unlikely to reflect true site-wide conversion rates, which are typically much lower; recall and precision should be re-checked once the model is scored against a more realistic, imbalanced population.
- Feature importance was derived from the Random Forest, not the deployed Logistic Regression, so the ranked driver list should be treated as directionally correct but not a literal explanation of the production model's decisions.
- The model does not currently incorporate temporal or seasonal effects (e.g., holiday shopping spikes), which may materially shift purchase likelihood independent of the behavioral features considered here.
- As with any behavioral targeting model, teams should ensure that personalization and discount-targeting use cases comply with applicable data-privacy and consumer-protection regulations in each market where the model is deployed.

8. Conclusion

This analysis demonstrates that a relatively simple, well-tuned Logistic Regression model can predict e-commerce purchase likelihood with strong accuracy (84.5%) and a high ROC-AUC (0.922), using a modest set of behavioral and demographic features. The clearest actionable finding is that purchase history and cart activity dominate the prediction, far outweighing demographics or acquisition channel — a result that should reorient marketing, personalization, and retention strategy toward behavior-first segmentation. With the recommended corrections to the ROC-AUC calculation, a phased rollout of the scoring model, and targeted pilots in cart-recovery and discount-targeting, the business is well positioned to convert this analysis into measurable revenue impact